Ranking Flaky Test Failures

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***Abstract*—Flaky tests pass and fail on the same code. Which failure is flaky? The ranking answers this question.**

I. Introduction

Flaky tests pass and fail on the same code [1]. Detectors rerun each failure [1], [2], and a survey covers the field [3]. Section II describes the method, Fig. 1 gives an overview, Table I lists the results, and Eq. (1) defines the score, also written as (1). Algorithm 1 ranks the failures, RQ1 asks how accurate the ranking is, and section III and section II close the paper on page 1.

A developer asked, "Why?? The test passed yesterday!" Another wrote "Really?!" and a third asked, "Is this the right page??" The value $f(?)$ stays open, and the most common (?) cause is the order of the tests. Is the cause known? Not yet. The ternary form `a ? b : c` and the query `?q=flaky` appear in the logs.

1) **RQ1:** How accurate is the ranking? (**RQ1**)

$$s(t) = \frac{f(t)}{r(t)} \quad (?) \tag{1}$$

II. Method

What now?? The tool ranks the failures. **Is it flaky?** The history decides.

```
grep -E "[?]" log.txt   # a literal ?? or (?) in the log
```

```
var name = user?.Name ?? "unknown";
```

TABLE I
Results

Project	Accuracy	Flaky?
A	0.71	yes
B	?	(?)



Fig. 1. Overview

Algorithm 1 Rank

rank the failures

III. Discussion

Which failures need attention first? The ranking answers this question for 71% of the builds.

References

[1] Q. Luo, F. Hariri, L. Eloussi, and D. Marinov, "An empirical analysis of flaky tests," in *Proceedings of the 22nd ACM SIGSOFT International Symposium on Foundations of Software Engineering*, 2014, pp. 643–653.

[2] J. Bell, O. Legunsen, M. Hilton, L. Eloussi, T. Yung, and D. Marinov, "DeFlaker: Automatically detecting flaky tests," in *Proceedings of the 40th International Conference on Software Engineering*, 2018, pp. 433–444.

[3] O. Parry, G. M. Kapfhammer, M. Hilton, and P. McMinn, "A survey of flaky tests," *ACM Transactions on Software Engineering and Methodology*, vol. 31, no. 1, pp. 1–74, 2021.